

INSTITUT UNIVERSITAIRE DES SCIENCES

Analyse du Trafic Réseau avec Wireshark

TD 1 – Réseaux II

Nom : EXUME

Prénom : Billy Rolph

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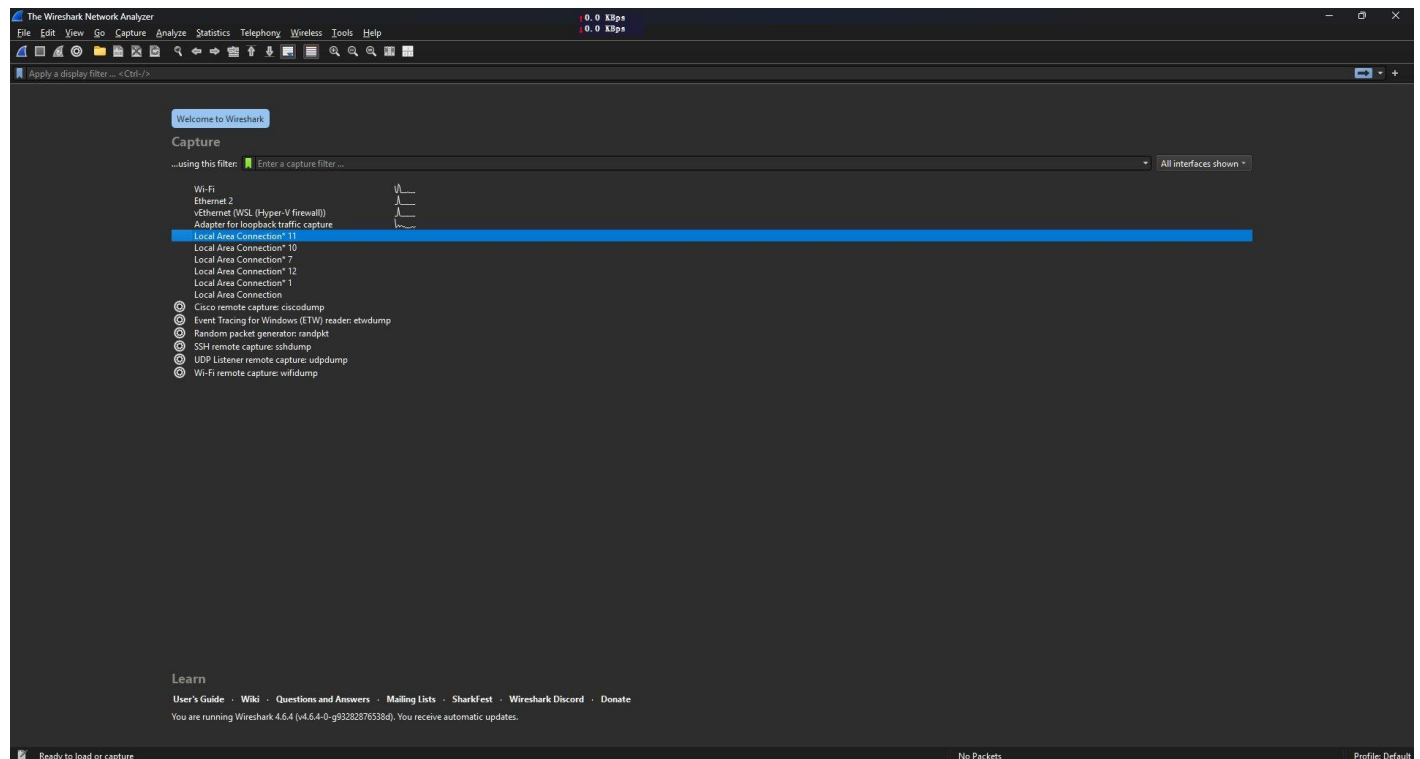
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Date : 22/03/2026

Introduction

Ce travail dirigé a pour objectif de découvrir et utiliser Wireshark, un outil d'analyse du trafic réseau. Nous avons capturé des paquets sur notre interface Wi-Fi, appliqué différents filtres pour isoler les protocoles ICMP, DNS, TCP et HTTP, puis utilisé les outils statistiques de Wireshark pour visualiser et interpréter le trafic. Les captures d'écran ci-dessous illustrent les différentes étapes réalisées.

Interface principale de Wireshark - Liste des interfaces réseau



Capture en cours - Paquets réseau qui défilent en temps réel

Wireshark interface showing a live capture of network traffic. The top pane displays a list of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The middle pane shows the details of the selected packet (No. 257), including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The bottom pane shows the raw packet data in hexadecimal and ASCII. The status bar at the bottom indicates 257 packets captured.

CMD - Commande ping google.com

```
C:\WINDOWS\system32\cmd. X + v

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rbill>ping google.com

Pinging google.com [142.250.176.78] with 32 bytes of data:
Reply from 142.250.176.78: bytes=32 time=45ms TTL=111
Reply from 142.250.176.78: bytes=32 time=64ms TTL=111
Reply from 142.250.176.78: bytes=32 time=71ms TTL=111
Reply from 142.250.176.78: bytes=32 time=63ms TTL=111

Ping statistics for 142.250.176.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 45ms, Maximum = 71ms, Average = 60ms

C:\Users\rbill>
```

Filtre ICMP - Paquets Echo Request et Echo Reply

The image shows a Wireshark network traffic capture. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The status bar at the top right shows 0.0 Kbps. The packet list pane on the left shows a list of captured packets, with packet 157 selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (ICMP). The packet bytes pane at the bottom shows the raw data of the selected packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
150	27.621988600	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=11/2016, ttl=128 (reply in 151)
151	27.667434300	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=11/2016, ttl=111 (request in 150)
157	28.642292600	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 159)
159	28.706515100	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=12/3072, ttl=111 (request in 157)
160	29.655669500	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=13/3328, ttl=128 (reply in 161)
161	29.726676300	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=13/3328, ttl=111 (request in 160)
165	30.683570100	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=14/3584, ttl=128 (reply in 166)
166	30.746431200	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=14/3584, ttl=111 (request in 165)

Frame 157: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{58576910-092F-460F-86AB-15C76E0923A4}, id 0
Ethernet II, Src: Intel_Z913187 (70:d8:23:79:13:87), Dst: ba:fd:e9:22:ad:92 (ba:fd:e9:22:ad:92)
Internet Protocol Version 4, Src: 10.56.202.126, Dst: 142.250.176.78
Internet Control Message Protocol

0000 ba fd e9 22 ad 92 70 d8 23 79 13 87 08 00 45 00 ... p #y... E
0010 00 3c 85 26 00 00 00 01 a1 9b 0a 38 ca 7e be fa ... & ... 8 ...
0020 b0 4e 08 00 4d 4f 00 01 00 0c 61 62 63 64 65 66 ... N MO ... abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ... ghijklm opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 ... wabcdefg hi

Détails d'un paquet ICMP sélectionné

The image shows a Wireshark network traffic capture with a packet details pane open for a selected ICMP Echo Request packet. The packet list pane on the left shows a list of captured packets, with packet 157 selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol (ICMP). The packet bytes pane at the bottom shows the raw data of the selected packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
150	27.621988600	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=11/2016, ttl=128 (reply in 151)
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157	28.642292600	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 159)
159	28.706515100	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=12/3072, ttl=111 (request in 157)
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161	29.726676300	142.250.176.78	10.56.202.126	ICMP	78	Echo (ping) reply id=0x0001, seq=13/3328, ttl=111 (request in 160)
165	30.683570100	10.56.202.126	142.250.176.78	ICMP	74	Echo (ping) request id=0x0001, seq=14/3584, ttl=128 (reply in 166)
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Frame 157: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{58576910-092F-460F-86AB-15C76E0923A4}, id 0
Ethernet II, Src: Intel_Z913187 (70:d8:23:79:13:87), Dst: ba:fd:e9:22:ad:92 (ba:fd:e9:22:ad:92)
Internet Protocol Version 4, Src: 10.56.202.126, Dst: 142.250.176.78
Internet Control Message Protocol

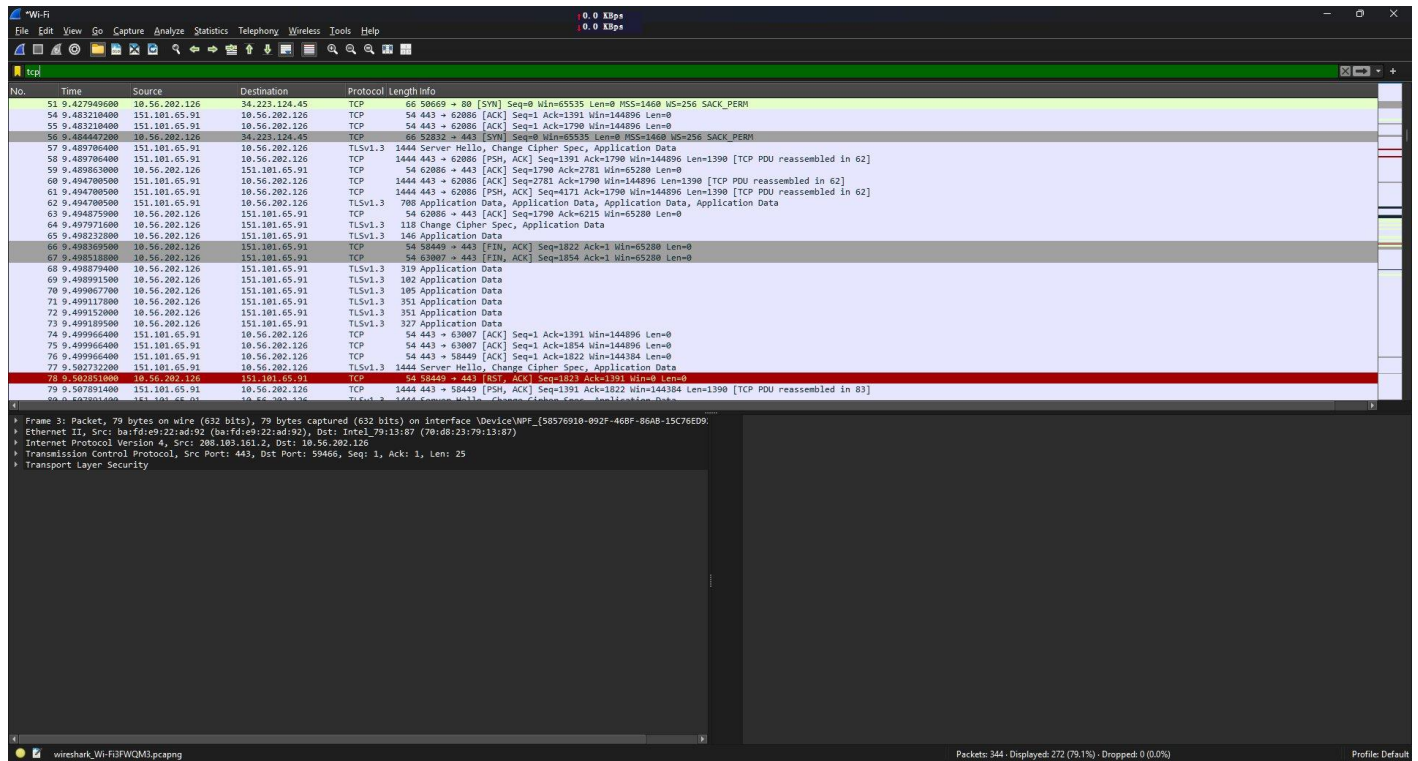
0000 ba fd e9 22 ad 92 70 d8 23 79 13 87 08 00 45 00 ... p #y... E
0010 00 3c 85 26 00 00 00 01 a1 9b 0a 38 ca 7e be fa ... & ... 8 ...
0020 b0 4e 08 00 4d 4f 00 01 00 0c 61 62 63 64 65 66 ... N MO ... abcdef
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ... ghijklm opqrstuv
0040 77 61 62 63 64 65 66 67 68 69 ... wabcdefg hi

No: 157 Time: 28.642292600 Source: 10.56.202.126 Destination: 142.250.176.78 Protocol: ICMP Length: 74 Info: Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 159)

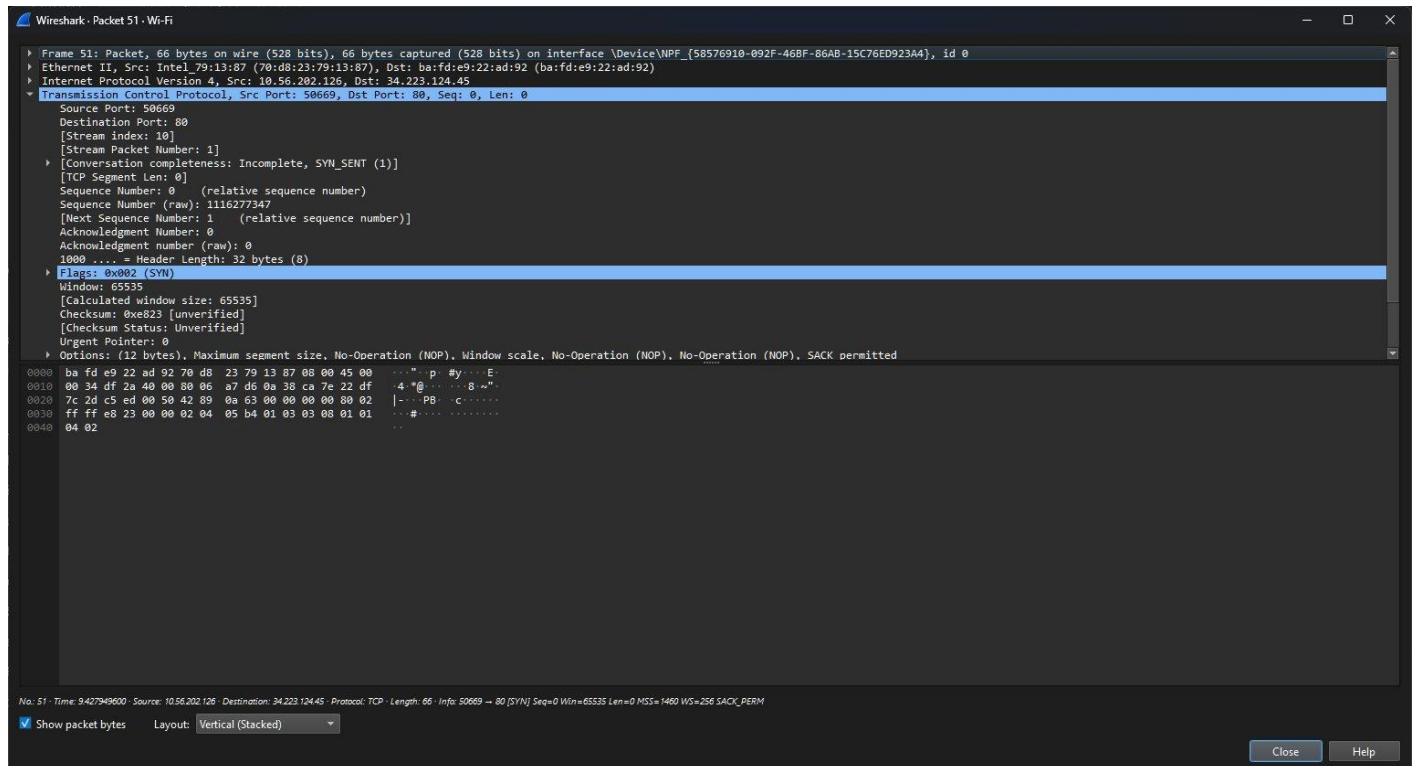
Show packet bytes Layout: Vertical (Stacked)

Close Help

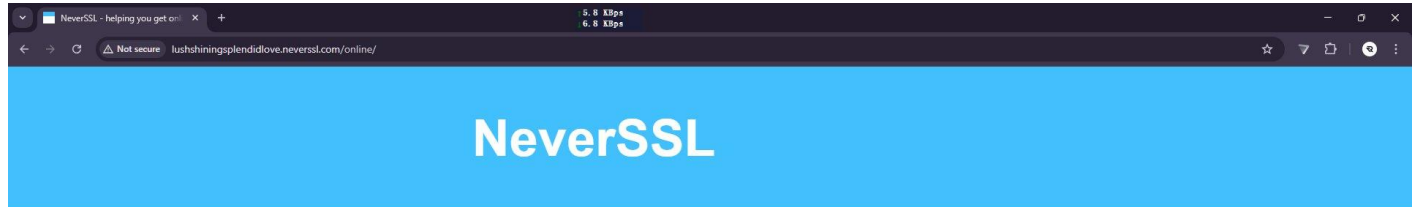
Filtre TCP - Handshake SYN / SYN-ACK / ACK visible



Détails du paquet SYN - Ports source et destination



Navigation sur NeverSSL - Site HTTP non chiffré



What?

This website is for when you try to open Facebook, Google, Amazon, etc on a wifi network, and nothing happens. Type "http://neverssl.com" into your browser's url bar, and you'll be able to log on.

How?

neverssl.com will never use SSL (also known as TLS). No encryption, no strong authentication, no [HSTS](#), no HTTP/2.0, just plain old unencrypted HTTP and forever stuck in the dark ages of internet security.

Why?

Normally, that's a bad idea. You should always use SSL and secure encryption when possible. In fact, it's such a bad idea that most websites are now using https by default.

And that's great, but it also means that if you're relying on poorly-behaved wifi networks, it can be hard to get online. Secure browsers and websites using https make it impossible for those wifi networks to send you to a login or payment page. Basically, those networks can't tap into your connection just like attackers can't. Modern browsers are so good that they can remember when a website supports encryption and even if you type in the website name, they'll use https.

And if the network never redirects you to this page, well as you can see, you're not missing much.

[Follow @neverssl](#)

Filtre HTTP - Requête GET et réponse 200 OK

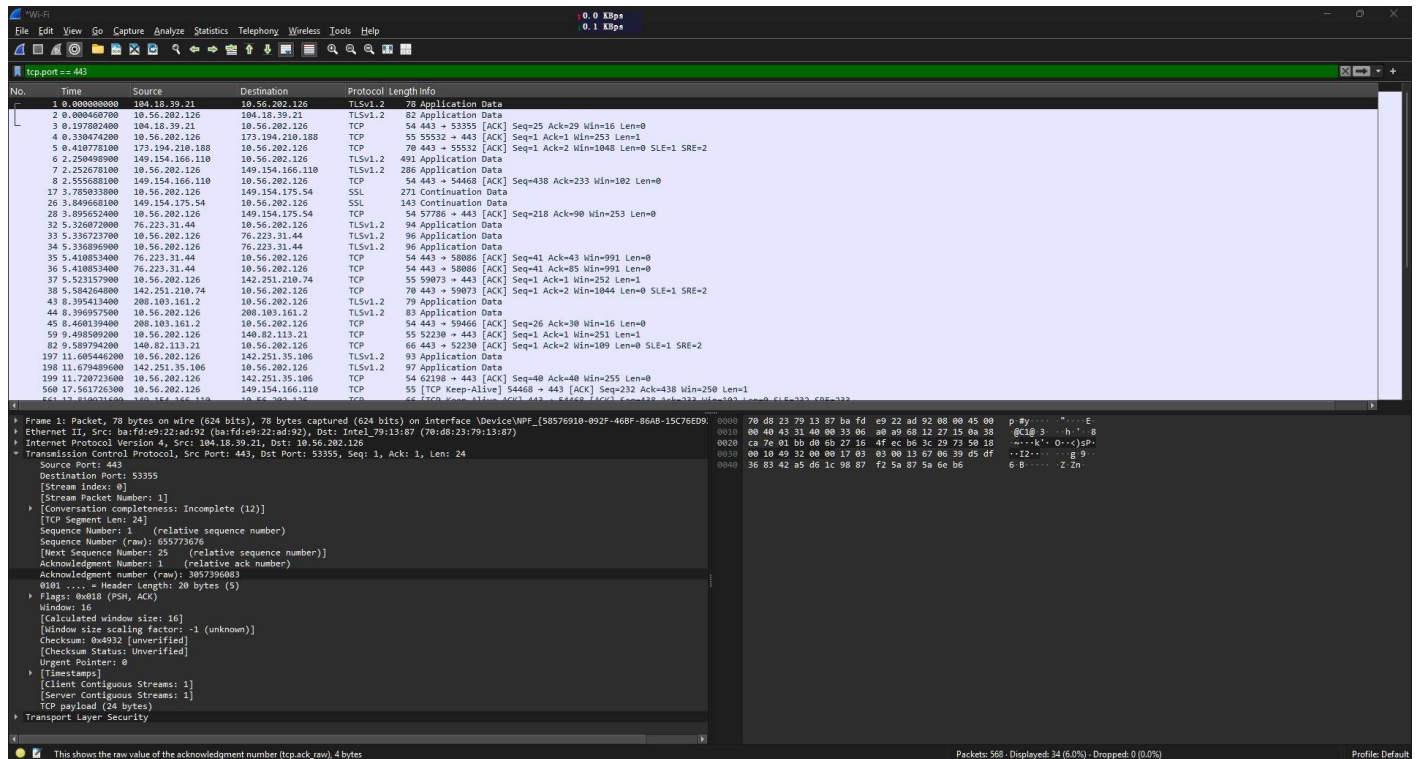
Wireshark network traffic analysis showing a GET request and a 200 OK response. The packet list shows a GET request from 10.56.202.126 to 10.56.202.126, and a 200 OK response from 10.56.202.126 to 10.56.202.126. The packet details pane shows the request and response structure, including the status bar at the bottom indicating 'Packets: 344, Displayed: 6 (1.7%), Dropped: 0 (0.0%)'.

No.	Time	Source	Destination	Protocol	Length	Info
162	12.595891500	10.56.202.126	34.223.124.45	HTTP	542	GET /online HTTP/1.1
165	12.771216300	34.223.124.45	10.56.202.126	HTTP	635	HTTP/1.1 301 Moved Permanently (text/html)
170	12.777320900	10.56.202.126	34.223.124.45	HTTP	543	GET /online/ HTTP/1.1
181	12.940410100	34.223.124.45	10.56.202.126	HTTP	273	HTTP/1.1 200 OK (text/html)
187	12.979985300	10.56.202.126	34.223.124.45	HTTP	477	GET /favicon.ico HTTP/1.1
199	13.137862100	34.223.124.45	10.56.202.126	HTTP	470	HTTP/1.1 200 OK (PNG)

Packet 165 details:

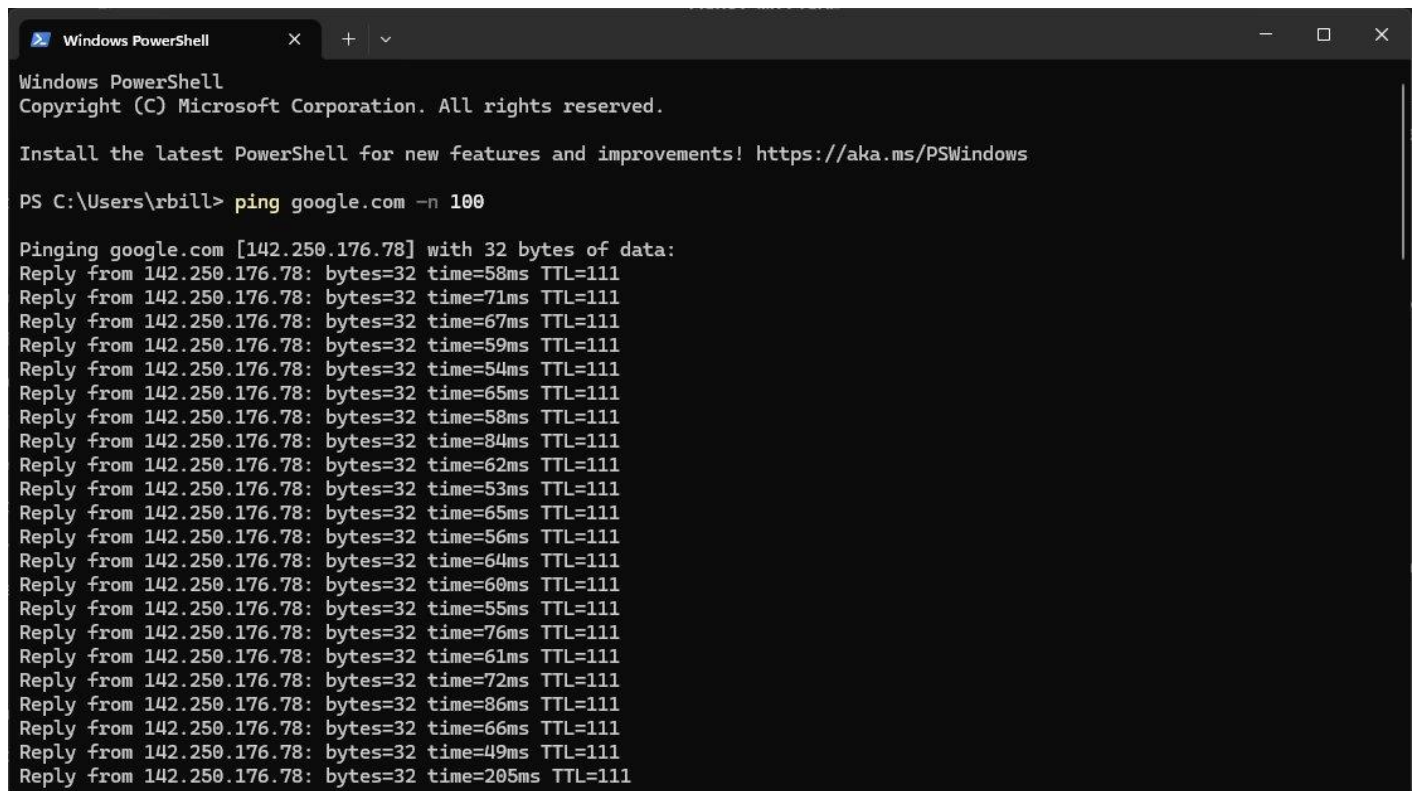
- IP: 10.56.202.126 → 10.56.202.126
- TCP: 54280 → 54280
- HTTP: GET /online HTTP/1.1
- Status: 200 OK

Filtre tcp.port == 443 - Trafic HTTPS chiffré (TLS)



Wireshark capture showing TLS traffic filtered by tcp.port == 443. The packet list shows various TLS records including Application Data, Handshake, and Alert. The packet details pane shows the structure of a TLS record, including the TLS header and the encrypted payload.

PowerShell - Commande ping google.com -n 100 (simulation ping flood)



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\rbill> ping google.com -n 100

Pinging google.com [142.250.176.78] with 32 bytes of data:
Reply from 142.250.176.78: bytes=32 time=58ms TTL=111
Reply from 142.250.176.78: bytes=32 time=71ms TTL=111
Reply from 142.250.176.78: bytes=32 time=67ms TTL=111
Reply from 142.250.176.78: bytes=32 time=59ms TTL=111
Reply from 142.250.176.78: bytes=32 time=54ms TTL=111
Reply from 142.250.176.78: bytes=32 time=65ms TTL=111
Reply from 142.250.176.78: bytes=32 time=58ms TTL=111
Reply from 142.250.176.78: bytes=32 time=84ms TTL=111
Reply from 142.250.176.78: bytes=32 time=62ms TTL=111
Reply from 142.250.176.78: bytes=32 time=53ms TTL=111
Reply from 142.250.176.78: bytes=32 time=65ms TTL=111
Reply from 142.250.176.78: bytes=32 time=56ms TTL=111
Reply from 142.250.176.78: bytes=32 time=64ms TTL=111
Reply from 142.250.176.78: bytes=32 time=60ms TTL=111
Reply from 142.250.176.78: bytes=32 time=55ms TTL=111
Reply from 142.250.176.78: bytes=32 time=76ms TTL=111
Reply from 142.250.176.78: bytes=32 time=61ms TTL=111
Reply from 142.250.176.78: bytes=32 time=72ms TTL=111
Reply from 142.250.176.78: bytes=32 time=86ms TTL=111
Reply from 142.250.176.78: bytes=32 time=66ms TTL=111
Reply from 142.250.176.78: bytes=32 time=49ms TTL=111
Reply from 142.250.176.78: bytes=32 time=205ms TTL=111
```


Statistics > Protocol Hierarchy - Proportion élevée d'ICMP (comportement suspect)

Wireshark - Protocol Hierarchy Statistics - Wi-Fi

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	908	100.0	158130	9213	0	0	0	908
Ethernet	100.0	908	8.0	12712	740	0	0	0	908
Internet Protocol Version 4	98.5	884	11.3	17880	1041	0	0	0	884
User Datagram Protocol	30.7	279	1.4	2232	130	0	0	0	279
QUIC IETF	24.3	221	29.0	45829	2670	221	42712	2488	229
Domain Name System	3.0	27	2.0	3233	188	27	3233	188	27
Data	3.4	31	0.7	1029	59	31	1029	59	31
Transmission Control Protocol	45.4	412	5.6	8888	517	255	5748	334	412
Transport Layer Security	16.0	145	28.8	45514	2651	145	40635	2367	146
Data	1.3	12	0.0	12	0	12	12	0	12
Internet Control Message Protocol	22.0	200	5.1	8000	466	200	8000	466	200
Data	0.3	3	2.7	4200	244	3	4200	244	3
Address Resolution Protocol	1.5	14	0.2	392	22	14	392	22	14
802.1Q Virtual LAN	19.7	179	0.5	716	41	0	0	0	179

No display filter.

Close Copy Protocols Help

Statistics > Conversations - Identification des IP avec le plus de paquets

Wireshark - Conversations - Wi-Fi

Conversation Settings

☐ Name resolution

☒ Absolute start time

☒ Display raw data

☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

Bluetooth

BPv7

DCCP

DNP 3.0

☒ Ethernet

FC

FDDI

IEEE 802.11

IEEE 802.15.4

ILNP

☒ IPv4

☒ IPv6

IPX

JXTA

LTP

MPTCP

NCP

openSAFETY

RSVP

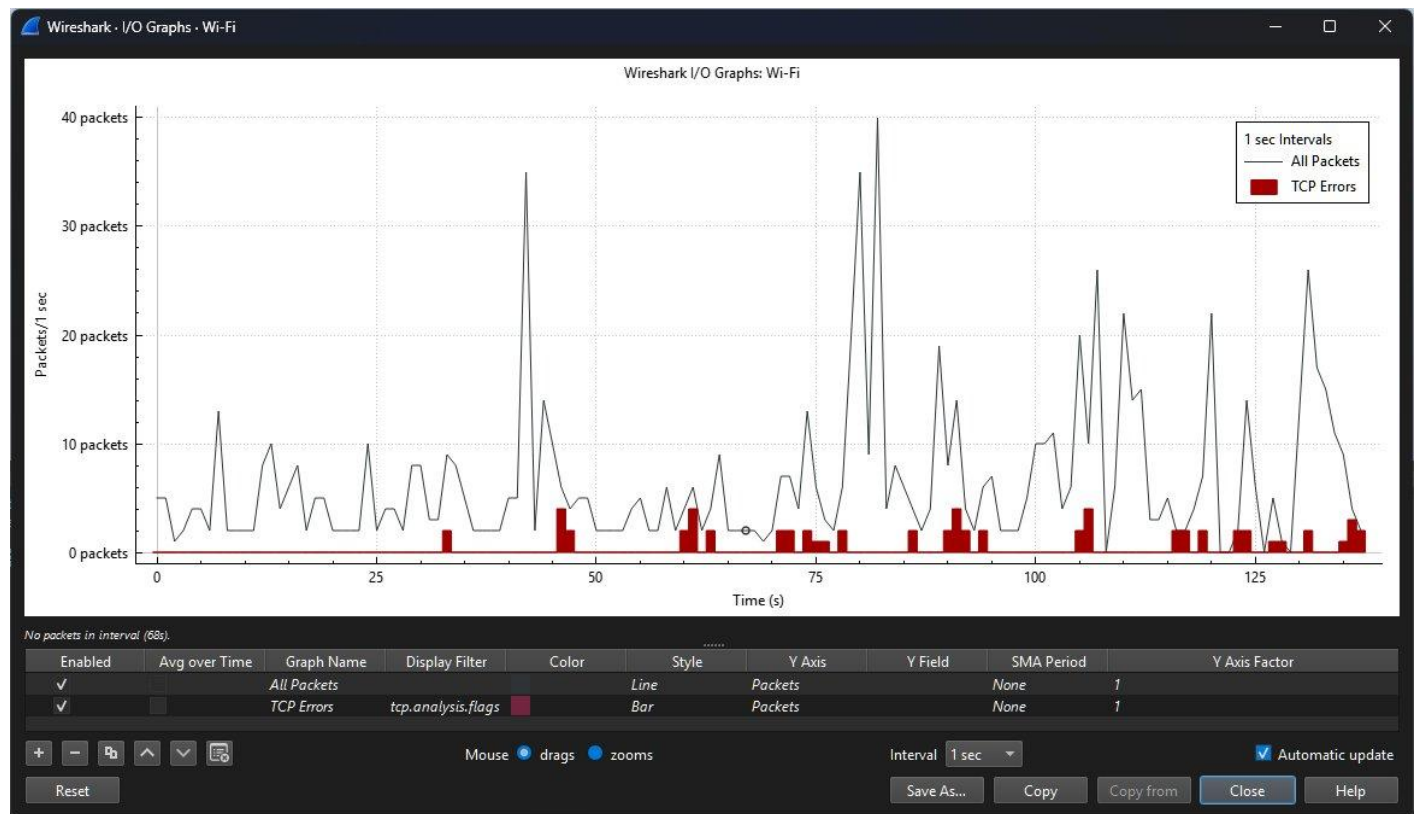
Filter list for specific type

Ethernet · 1	IPv4 · 33	IPv6	TCP · 30	UDP · 17								
Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.56.202.126	10.56.202.9	27	4 kB	6	14	1 kB	13	3 kB	5.688243000	114.814909	82 bits/s	221 bits/s
10.56.202.126	13.89.179.14	3	162 bytes	0	2	108 bytes	1	54 bytes	0.000000000	0.163590	5281 bits/s	2640 bits/s
10.56.202.126	18.97.36.54	17	2 kB	20	8	1 kB	9	864 bytes	29.211410500	90.362635	116 bits/s	76 bits/s
10.56.202.126	20.189.173.23	40	15 kB	30	20	7 kB	20	8 kB	82.081610300	51.498896	1028 bits/s	1276 bits/s
10.56.202.126	23.47.204.32	4	240 bytes	10	2	108 bytes	2	132 bytes	13.223047300	0.071640	12 kbps	14 kbps
10.56.202.126	23.221.212.132	28	8 kB	25	13	1 kB	15	7 kB	42.020753200	0.279212	42 kbps	199 kbps
10.56.202.126	23.221.212.136	4	240 bytes	11	2	108 bytes	2	132 bytes	13.223302500	0.059393	14 kbps	17 kbps
10.56.202.126	34.54.84.110	16	1 kB	14	7	549 bytes	9	712 bytes	16.027101500	120.026565	36 bits/s	47 bits/s
10.56.202.126	52.123.247.63	13	7 kB	31	7	2 kB	6	5 kB	89.764792800	0.560213	24 kbps	76 kbps
10.56.202.126	52.179.73.36	18	8 kB	32	8	3 kB	10	5 kB	120.418228500	0.475292	43 kbps	92 kbps
10.56.202.126	57.144.23.32	65	19 kB	8	27	3 kB	38	16 kB	7.164526600	120.333703	198 bits/s	1049 bits/s
10.56.202.126	76.223.31.44	6	363 bytes	24	3	165 bytes	3	198 bytes	41.418821800	90.177525	14 bits/s	17 bits/s
10.56.202.126	135.234.174.40	6	363 bytes	15	3	165 bytes	3	198 bytes	16.294869900	90.289064	14 bits/s	17 bits/s
10.56.202.126	140.82.112.25	10	698 bytes	22	4	276 bytes	6	422 bytes	30.882669000	74.967882	29 bits/s	45 bits/s
10.56.202.126	142.250.176.78	200	15 kB	7	100	7 kB	100	8 kB	5.800905700	100.757216	587 bits/s	619 bits/s
10.56.202.126	142.251.35.106	21	2 kB	17	12	1 kB	9	949 bytes	19.722376800	105.008893	79 bits/s	72 bits/s
10.56.202.126	142.251.107.188	6	375 bytes	21	3	165 bytes	3	210 bytes	29.594204000	90.251560	14 bits/s	18 bits/s
10.56.202.126	142.251.152.119	3	253 bytes	1	1	71 bytes	2	182 bytes	0.387142400	23.934080	23 bits/s	60 bits/s
10.56.202.126	142.251.210.74	10	672 bytes	3	5	273 bytes	5	399 bytes	1.303997200	104.904246	20 bits/s	30 bits/s
10.56.202.126	142.251.214.170	6	375 bytes	12	3	165 bytes	3	210 bytes	15.571071000	90.192665	14 bits/s	18 bits/s
10.56.202.126	149.154.166.110	22	4 kB	4	9	1 kB	13	3 kB	3.198597100	121.278891	93 bits/s	165 bits/s
10.56.202.126	149.154.175.54	20	3 kB	26	12	2 kB	8	966 bytes	44.550273900	90.107712	177 bits/s	85 bits/s
10.56.202.126	172.172.255.217	3	434 bytes	27	2	209 bytes	1	225 bytes	73.206620300	0.141008	11 kbps	12 kbps
10.56.202.126	172.172.255.218	3	379 bytes	23	2	151 bytes	1	228 bytes	40.002790200	0.152539	7919 bits/s	11 kbps
10.56.202.126	173.194.210.188	12	750 bytes	19	6	330 bytes	6	420 bytes	27.418116300	95.794388	27 bits/s	35 bits/s
10.56.202.126	184.28.118.34	184	35 kB	28	86	15 kB	98	20 kB	79.791724600	56.275297	2145 bits/s	2842 bits/s
10.56.202.126	185.199.108.133	6	363 bytes	16	3	165 bytes	3	198 bytes	16.342865900	90.184191	14 bits/s	17 bits/s
10.56.202.126	185.199.109.154	16	968 bytes	2	8	440 bytes	8	528 bytes	1.256446700	136.051903	25 bits/s	31 bits/s
10.56.202.126	185.199.110.133	6	363 bytes	18	3	165 bytes	3	198 bytes	26.363808300	90.216293	14 bits/s	17 bits/s
10.56.202.126	192.178.50.46	18	10 kB	29	8	5 kB	10	5 kB	80.035435300	0.335798	116 kbps	119 kbps
172.64.148.235	10.56.202.126	21	1 kB	13	14	924 bytes	7	574 bytes	15.671161700	120.073558	61 bits/s	38 bits/s
208.103.161.2	10.56.202.126	30	2 kB	9	20	1 kB	10	975 bytes	12.144469800	120.320178	98 bits/s	64 bits/s
216.239.32.223	10.56.202.126	50	12 kB	5	26	7 kB	24	6 kB	4.573083200	129.990743	417 bits/s	349 bits/s

Close

Help

Statistics > I/O Graph - Pic de trafic ICMP visualisé sur le graphique



Conclusion

Ce TD nous a permis de prendre en main Wireshark et de comprendre les échanges réseau à travers l'analyse des protocoles ICMP, DNS, TCP et HTTP. Nous avons pu observer le handshake TCP en trois étapes, identifier des requêtes DNS, analyser du trafic HTTP en clair et détecter un comportement suspect grâce aux statistiques. Wireshark est un outil essentiel pour le diagnostic réseau et la détection d'anomalies.